Notes on EE Teaching, Fourier Transforms, and Schwartz Spaces

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1 The experience that prompted the discussion

I learned Fourier transforms primarily through engineering, particularly electrical engineering, during 1982–1986, rather than through mathematics courses. I began graduate school in 1990. As I recall, I learned about Banach spaces and dual spaces around 1992, and a little about Sobolev spaces around 1994. These dates are approximate.

A few years later, thinking about dual spaces led me to an idea: start with the Fourier transform on a suitable space of smooth functions contained in L^1, then use its adjoint, or dual operator, to extend the transform to a larger class of objects. Only last month did I learn about Schwartz spaces and tempered distributions, which provide a systematic realization of this idea.

This history made me wonder how much of that framework my undergraduate EE teachers knew. Fourier transforms were central to their teaching, yet the mathematical foundation for transforming nonintegrable signals and using Dirac impulses was not made clear to me. As an undergraduate engineer, I was frustrated because the Dirac delta function seemed fictitious.

2 What was wrong with the apparent delta "function"?

That objection has a precise mathematical basis. No Lebesgue-integrable ordinary function that vanishes everywhere except at zero can have integral one: changing a function at a single point does not change its integral. Describing the delta as zero away from zero, infinitely tall at zero, and having unit area therefore does not define an ordinary integrable function.

The distributional formulation changes the type of object under discussion. For a smooth test function φ, define

$$\langle \delta, \varphi \rangle = \varphi(0).$$

The delta is a continuous linear functional on an appropriate test-function space. An ordinary function f can also define a functional, through

$$\langle T_f, \varphi \rangle = \int_{\mathbb{R}} f(x)\varphi(x)\,dx,$$

whenever this pairing is well defined and continuous on the chosen space. For example, every locally integrable function defines a distribution on compactly supported smooth test functions. Additional conditions are needed for it to define a tempered distribution on Schwartz space.

The familiar narrow-pulse picture can be made precise without first teaching a full course in functional analysis. Set

$$p_\varepsilon(x) = \frac{1}{2\varepsilon}\mathbf{1}_{[-\varepsilon,\varepsilon]}(x), \qquad \varepsilon > 0.$$

For every smooth test function,

$$\langle T_{p_\varepsilon}, \varphi \rangle = \frac{1}{2\varepsilon} \int_{-\varepsilon}^{\varepsilon} \varphi(x)\, dx \longrightarrow \varphi(0) = \langle \delta, \varphi \rangle.$$

Indeed, the difference from $\varphi(0)$ is bounded by $\sup_{|x|\le\varepsilon} |\varphi(x) - \varphi(0)|$, which tends to zero. The convergence is convergence of actions on test functions. It is not convergence in L^1 to an ordinary delta function.

3 The dual-space construction of the Fourier transform

Fix the convention

$$(\mathcal{F}\varphi)(\xi) = \int_{\mathbb{R}} \varphi(x) e^{-2\pi i x\xi}\, dx, \qquad (\mathcal{F}^{-1}\psi)(x) = \int_{\mathbb{R}} \psi(\xi) e^{2\pi i x\xi}\, d\xi.$$

Schwartz space $\mathcal{S}(\mathbb{R})$ consists of smooth functions for which

$$p_{m,n}(\varphi) = \sup_{x\in\mathbb{R}} |x^m \varphi^{(n)}(x)| < \infty \quad \text{for all integers } m, n \ge 0.$$

These seminorms specify its topology, which is indeed metrizable. To give an explicit metric, combine them into the increasing family

$$q_k(\varphi) = \max_{0\le m,n\le k} p_{m,n}(\varphi), \qquad k = 0, 1, 2, \ldots,$$

and set

$$\boxed{d(\varphi, \psi) = \sum_{k=0}^{\infty} 2^{-(k+1)} \min\{1, q_k(\varphi - \psi)\}.}$$

This is a translation-invariant metric inducing exactly the Schwartz topology. Positivity follows because $q_0(\varphi - \psi) = 0$ forces equality; the triangle inequality follows from the seminorm inequalities and $\min\{1, a + b\} \le \min\{1, a\} + \min\{1, b\}$ for $a, b \ge 0$. Moreover,

$$d(\varphi_j, \varphi) \longrightarrow 0 \quad \Longleftrightarrow \quad p_{m,n}(\varphi_j - \varphi) \longrightarrow 0 \quad \text{for every } m, n.$$

For the forward implication, each fixed nonnegative summand must tend to zero. For the reverse implication, first make the geometric tail small, then make the finitely many remaining terms small.

This metric is complete. A metric-Cauchy sequence is Cauchy in every $p_{m,n}$; its derivatives converge uniformly, their limits are compatible under differentiation, and the weighted uniform limits give all the Schwartz bounds and convergence in each seminorm. Thus $\mathcal{S}$ is a Fréchet space: a complete metrizable locally convex vector space. The displayed metric is not a norm, and the Schwartz topology is not induced by any single norm.

The Fourier transform is a continuous isomorphism

$$\mathcal{F} : \mathcal{S}(\mathbb{R}) \longrightarrow \mathcal{S}(\mathbb{R}).$$

Its continuous linear dual $\mathcal{S}'(\mathbb{R})$ is the space of tempered distributions. Using a bilinear pairing, with no complex conjugation, define

$$\boxed{\langle \mathcal{F}T, \varphi \rangle = \langle T, \mathcal{F}\varphi \rangle, \qquad T \in \mathcal{S}', \quad \varphi \in \mathcal{S}.}$$

The inverse is defined analogously using $\mathcal{F}^{-1}$. This extends the ordinary Fourier transform: for $f \in L^1$, interchanging the absolutely convergent integrals gives

$$\langle T_{\mathcal{F}f}, \varphi \rangle = \langle T_f, \mathcal{F}\varphi \rangle.$$

In particular, with this normalization,

$$\mathcal{F}\delta = 1, \qquad \mathcal{F}1 = \delta, \qquad \mathcal{F}\big(e^{2\pi i a x}\big) = \delta_a, \quad \langle \delta_a, \varphi \rangle = \varphi(a).$$

These are identities of tempered distributions, not assertions that the ordinary Fourier integral converges absolutely for a constant or sinusoid.

The original adjoint idea therefore identified the essential mechanism. More precisely, this is the *transpose* acting on a continuous linear dual; it should be distinguished from a Hilbert-space adjoint defined using a sesquilinear inner product. The choice of test space and topology is a substantial part of the construction. Merely specifying "smooth functions in L^1" does not specify a topology or guarantee Fourier invariance.

Failure to belong to L^1 is not itself a failure of Fourier analysis. There is also the L^2 extension by the Plancherel theorem, distinct from the broader tempered-distribution setting. Sobolev spaces provide another connection: for real s, one can define $H^s(\mathbb{R})$ as the tempered distributions whose Fourier transforms are represented by functions satisfying

$$\int_{\mathbb{R}} (1 + 4\pi^2 \xi^2)^s |\widehat{T}(\xi)|^2 \, d\xi < \infty.$$

This illustrates why Sobolev spaces belong in the discussion, although knowing about them does not by itself establish knowledge of the full Schwartz-space construction.

4 Topology and sequential convergence of tempered distributions

The topology on the test-function space $\mathcal{S}$ and the topology on its dual $\mathcal{S}'$ are different constructions. There are two standard dual topologies worth distinguishing.

4.1 The weak-star topology: test one function at a time

The weak-star topology $\sigma(\mathcal{S}', \mathcal{S})$ is generated by the seminorms

$$r_\varphi(T) = |\langle T, \varphi \rangle|, \qquad \varphi \in \mathcal{S}.$$

A basic neighborhood of T imposes finitely many inequalities $|\langle U - T, \varphi_j \rangle| < \varepsilon_j$. Consequently, to prove that T_j converges to a given $T \in \mathcal{S}'$, it is necessary and sufficient to prove

$$\boxed{\langle T_j, \varphi \rangle \longrightarrow \langle T, \varphi \rangle \quad \text{for every fixed } \varphi \in \mathcal{S}.}$$

The test function is arbitrary but held fixed while j tends to infinity. This is often called weak convergence of tempered distributions; the term weak-star specifies precisely the pairing being used. See also [8], Section 3.3.3.

There is a useful existence version of this criterion. Suppose each $T_j \in \mathcal{S}'$ and $\langle T_j, \varphi \rangle$ has a finite limit for every $\varphi \in \mathcal{S}$. Define $T(\varphi)$ to be that limit. Linearity is immediate, but continuity needs justification. Pointwise convergence implies pointwise boundedness, and the Banach–Steinhaus theorem for the Fréchet space $\mathcal{S}$ gives common constants $C > 0$ and k such that

$$|\langle T_j, \varphi \rangle| \le C q_k(\varphi) \quad \text{for all } j \text{ and } \varphi \in \mathcal{S}.$$

Passing to the limit proves the same bound for T, so $T \in \mathcal{S}'$. Thus convergence of all these scalar pairings automatically produces a tempered-distribution limit.

In a direct proof, one can instead establish a common bound of this form and compute the limits of the pairings. Checking only a dense collection of test functions requires such uniform control to extend the conclusion to all of $\mathcal{S}$; density alone is insufficient.

4.2 The strong dual topology: uniformity on bounded test sets

A set $B \subset \mathcal{S}$ is bounded if $\sup_{\varphi \in B} q_k(\varphi) < \infty$ for every k. Here boundedness is in the locally convex sense, not boundedness in the displayed metric (that metric is bounded by one on the entire space). The strong dual topology $\beta(\mathcal{S}', \mathcal{S})$ is generated by

$$P_B(T) = \sup_{\varphi \in B} |\langle T, \varphi \rangle|, \qquad B \subset \mathcal{S} \text{ bounded.}$$

Strong convergence means $P_B(T_j - T) \to 0$ for every such B.

For *sequences* of tempered distributions, weak-star and strong convergence coincide. This is a special property of Schwartz space, not a general property of dual spaces. Here is the argument. If $T_j \to T$ pointwise on $\mathcal{S}$, Banach–Steinhaus makes $R_j = T_j - T$ equicontinuous:

$$|\langle R_j, \varphi \rangle| \le C q_k(\varphi) \quad \text{with fixed } C, k.$$

Schwartz space is a Montel space: every bounded subset has compact closure. This compactness follows from Arzelà–Ascoli on compact intervals, a diagonal extraction for all derivatives, and the higher weighted seminorm bounds controlling the tails at infinity.

Given a bounded set B and $\varepsilon > 0$, cover its compact closure by finitely many sets $q_k(\varphi - \varphi_\ell) < \varepsilon/(2C)$. For sufficiently large j, convergence at the finitely many centers gives $|\langle R_j, \varphi_\ell \rangle| < \varepsilon/2$ for every ℓ. For any $\varphi \in B$, choose one such center. Then

$$|\langle R_j, \varphi \rangle| \le C q_k(\varphi - \varphi_\ell) + |\langle R_j, \varphi_\ell \rangle| < \varepsilon.$$

This proves strong convergence. The reverse implication follows by taking $B = \{\varphi\}$.

The two topologies nevertheless differ on the full dual space; equality of convergent sequences does not imply equality of topologies here. Neither of these topologies on the full $\mathcal{S}'(\mathbb{R})$ is metrizable. The metric on $\mathcal{S}$ should therefore not be mistaken for a metric on its dual, and the sequential equivalence just proved should not be asserted for arbitrary nets.

4.3 A concrete convergence proof and its Fourier consequence

For the narrow pulses defined earlier, the fundamental theorem of calculus gives the explicit estimate

$$|\langle T_{p_\varepsilon} - \delta, \varphi \rangle| \le \frac{1}{2\varepsilon} \int_{-\varepsilon}^{\varepsilon} |x| \sup_y |\varphi'(y)| \, dx = \frac{\varepsilon}{2} p_{0,1}(\varphi).$$

It proves weak-star convergence immediately. Taking the supremum over any bounded $B \subset \mathcal{S}$ proves strong convergence directly as well. Thus every sequence $\varepsilon_j > 0$ tending to zero gives $T_{p_{\varepsilon_j}} \to \delta$ in both topologies.

Finally, if $T_j \to T$ in $\mathcal{S}'$, then for every $\varphi \in \mathcal{S}$,

$$\langle \mathcal{F} T_j, \varphi \rangle = \langle T_j, \mathcal{F}\varphi \rangle \longrightarrow \langle T, \mathcal{F}\varphi \rangle = \langle \mathcal{F} T, \varphi \rangle.$$

This proves convergence of the Fourier transforms in $\mathcal{S}'$. More generally, the Fourier transform is continuous for both dual topologies: for the strong topology, use the fact that $\mathcal{F}$ maps bounded subsets of $\mathcal{S}$ to bounded subsets of $\mathcal{S}$.

5 What can be inferred about the teachers?

Three different levels of knowledge should be distinguished:

1. Using impulses, impulse trains, and sinusoidal Fourier transforms correctly in engineering calculations.
2. Knowing that distribution theory provides a rigorous foundation.
3. Understanding the test spaces, their topologies, their continuous duals, and the extension of the Fourier transform by transposition.

Expertise at the first level does not prove knowledge at the third. Conversely, omission from an undergraduate course does not prove that the instructor lacked that knowledge. An instructor might know the framework, know only that it exists, or have little familiarity with it.

The initial suggestion that the teachers were "stuck in the 1930s" was stronger than the evidence supported. Schwartz's systematic distribution theory dates to the 1940s, so the foundation existed well before the undergraduate period in question [1]. Nevertheless, the age of the foundations presented in a course does not date an instructor's expertise in applications or other parts of electrical engineering.

An initial counterexample in the discussion was Oppenheim's MIT course, developed in 1987, which included Fourier representations of periodic signals [2, 3]. That establishes use of generalized-transform ideas in a practical sense. It does *not* establish that the course taught Schwartz-space duality, and therefore did not adequately answer the more specific question.

6 A limited check of graduate EE teaching

To make the inquiry more concrete, a professor was randomly selected from a manually assembled list of 22 professors and associate professors in Penn State's EE directory. This was not a statistically representative survey of the whole faculty. The randomly chosen professor's doctoral institution was Penn State itself, so the inquiry led back to Penn State.

The search located Asok Ray's Fall 2023 syllabus for EE/ME 550, *Foundations of Engineering Systems Analysis*, a first-year graduate course [4]. It explicitly includes function spaces, Fourier theory, duality, adjoint spaces, representation theorems, and spectral analysis. The following findings concern that documented syllabus and the book material inspected, not the randomly chosen professor's personal coursework or the textbooks used at Penn State in the early 1980s.

6.1 Naylor and Sell: the assigned textbook

The assigned text is A. W. Naylor and G. R. Sell, *Linear Operator Theory in Engineering and Science* (1982). Springer describes it as intended for engineers, scientists, and applied mathematicians; it was first published in 1971 [5]. Inspection of its publisher-provided index and bibliography established the following [6]:

- Algebraic conjugate spaces are indexed at pp. 208–212 and normed conjugate spaces at p. 270.
- Smooth functions and compactly supported smooth functions appear in the index of symbols.
- Fourier-transform theory is indexed at pp. 334–335 and 360–366.

- Sobolev spaces are indexed at pp. 281–282, with further references in the partial differential equation material.
- Distribution theory is indexed at p. 222, and the bibliography includes Schwartz's *Théorie des distributions*.
- No index entries for Schwartz space or tempered distributions were found in the inspected index.

This confirms substantial functional analysis and Sobolev-space content in an engineering-oriented textbook actually assigned in a graduate EE course. It does not verify the particular construction $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ extended to $\mathcal{S}'$ by duality. Separate index entries for dual spaces and Fourier transforms do not establish that connection. Nor does absence from an index prove absence from the full text.

6.2 Bressan: a listed reference book

The same syllabus lists Alberto Bressan's *Lecture Notes on Functional Analysis: With Applications to Linear Partial Differential Equations* (2013) as a reference, rather than the assigned textbook [4]. Its published contents include distributions and weak derivatives, Sobolev spaces, adjoint operators, and Hilbert spaces [7]. The inquiry did not verify a treatment of Schwartz spaces and tempered-distribution Fourier transforms. It is a mathematics book recommended in an EE course; that is the precise connection established here.

I know Alberto Bressan personally, although I have not spoken to him in several years. This personal connection is part of the discussion's context, not additional evidence about what EE students or professors learned.

6.3 Limits of the evidence

The inquiry yielded a partial affirmative answer: substantial functional analysis, Sobolev spaces, and distribution theory do appear in the documented graduate EE reading material. It did not produce a verified list of Penn State EE course books covering the entire Schwartz-space Fourier framework. The 2023 syllabus cannot establish the content of a particular professor's 1980s training, and an available reference book does not establish that students read the relevant chapters.

On this limited evidence, it seems quite possible that at least some EE professors know about generalized Fourier transforms at a rigorous level. It also remains plausible that some complete advanced EE training without learning the particular dual-space construction discussed here. No numerical claim about how many teachers knew it is justified by this inquiry.

7 The teaching concern that remains

Fourier transforms are central to electrical engineering, and the examples that prompted the concern include constants, everlasting sinusoids, and Dirac impulses. It is reasonable for a student to ask what kinds of objects these are, what the transform returns, and in what sense the equations hold. Stating an L^1 condition for an ordinary integral and then transforming these examples without explanation leaves a genuine conceptual gap.

An undergraduate course need not develop the full topology of Schwartz space to address that gap. It can say that the delta is a functional rather than an ordinary function, explain its action on test functions, demonstrate the narrow-pulse limit, and identify the resulting Fourier identities as distributional identities. A reference can point interested students to the complete theory.

The strongest criticism supported by the discussion is therefore about the foundation communicated to the student. Omitting the full machinery can be a sensible teaching decision. Omitting even an acknowledgment of the mathematical change of setting can leave a careful student thinking that an essential tool is fictitious. A short, precise explanation could have addressed that frustration long before a course on dual spaces.

References

[1] MacTutor History of Mathematics, Laurent Schwartz: biography.

[2] MIT OpenCourseWare, Signals and Systems, course information. The course description dates its development to 1987.

[3] MIT OpenCourseWare, Lecture 8: Continuous-Time Fourier Transform.

[4] Asok Ray, Penn State, EE/ME 550 syllabus, Fall 2023.

[5] A. W. Naylor and G. R. Sell, Linear Operator Theory in Engineering and Science, Springer, 1982; publisher's description and contents.

[6] Naylor and Sell, Back matter, especially the bibliography and indexes, pp. 609–624.

[7] Universities Press / Orient BlackSwan, Mathematics catalogue, p. 60: contents of Alberto Bressan's *Lecture Notes on Functional Analysis*.

[8] Stéphane Nonnenmacher, Distributions and PDEs, Fall 2022 lecture notes, Chapter 3, especially Section 3.3.3.